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SUPERSEDING
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MILITARY SPECIFICATION

CLOTH, WIND RESISTANT OXFORD, NYLON AND COTTON, QUARPEL TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of nylon and cotton blended oxford cloth with a Quarpel type water repellent treatment.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 Materials (see 6.7).

3.2.1 Fibers.

3.2.1.1 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a normal cut length of 1-1/2 inches (38 mm) and a round cross section with a single nominal denier of 2.25 - 2.5 (.25 - .28 Tex). The contractor shall submit the fiber producer's certification that each lot of nylon staple used conforms to the requirements specified herein.

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3.2.1.2 Cotton. The cotton shall be carded.

3.2.2 Yarns. The yarns shall be a blend of nylon and cotton, drawn and spun into single yarns for both warp and filling. The finished cloth shall contain 50 ± 3 percent nylon and the remaining percentage cotton, based on the dry weight of the desized specimen. Tests shall be performed as specified in 4.2.3.

3.3 Color. The color shall be as specified (see 6.2 and 6.4). The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid shall be chosen and applied such that the dyed cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When no standard sample is available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

3.3.1 Matching. The finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7000 ± 500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 kelvin.

3.3.2 Colorfastness. The dyed and finished cloth shall show fastness to laundering (after three cycles), crocking, perspiration and light equal to or better than the standard sample when tested as specified in 4.2.3. When no standard sample has been established or designated as applicable to colorfastness, the dyed and finished cloth shall show "good" fastness to light, laundering (after three cycles), perspiration and crocking when tested as specified in 4.2.3.

3.4 Physical requirements. The finished cloth shall conform to the requirements in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Weight oz. per sq. yd. (g/m^2)		Tearing strength pounds (Newtons)	Yarns per inch (cm)		Breaking strength pounds (Newtons)		Weave
			Minimum				
Min.	Max.	Filling	Warp	Filling	Warp	Filling	
6.3 (214)	7.1 (241)	3.5 (16)	130 (51)	48 (19)	240 (1068)	90 (400)	Oxford

3.4.1 Width. The width of the cloth shall be as specified (see 6.2). The minimum acceptable width shall be inclusive of woven selvages when fly-shuttle looms are used, and exclusive of lock in selvages when shuttleless looms are used.

3.5 Finish. The cloth shall be singed, desized, scoured, mercerized, and dyed. The cloth shall be given a water repellent treatment as specified in 3.7.

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3.5.1 Nonfibrous material. The finished cloth, prior to the application of the water repellent treatment shall not exceed 2.0 percent starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.2.3.

3.6 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 3.5 percent in the warp direction nor more than 2.5 percent in the filling direction when tested as specified in 4.2.3. The preshrinking process used shall not be identified by name or trademark either on the cloth, ticket, or package.

3.7 Water repellency and air permeability. The cloth shall be given an approved Quarpel type (see 6.5) water repellent treatment and shall conform to the water repellency and air permeability requirements of 3.7.1 and table II when tested as specified in 4.2.3.

TABLE II. Water-repellency and air permeability requirements

	Hydrostatic height minimum		Dynamic absorption maximum		Air permeability
	Average cm.	Min. 1/ cm.	Average Percent	Max. 2/ Percent	cu.ft./ minute per sq. foot, ($cm^3/sec/cm^2$) max. (avg.)
Initial	30	25	20	25	-
After 5 launderings	30	25	20	25	4.5 (2.3)

1/ No single determination shall fall below the specified minimum.

2/ No single determination shall exceed the specified maximum.

3.7.1 Spray rating. The measurements of the three individual determinations on the sample unit for initial spray rating, when tested as specified in 4.2.3, shall be equal to or better than 90, 90, 80.

3.7.2 Resistance to organic liquid. The finished cloth shall not show wetting by n-tetradecane either initially or after 5 launderings when tested as specified in 4.2.3.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 6.5 nor more than 8.5 when tested as specified in 4.2.3.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.2.3.

3.10 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (37 m). The cloth shall be put up on full width rolls as specified in PPP-P-1134.

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3.11 Fiber identification. Each roll of cloth shall be labeled or ticketed, for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Face side identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (91 m^2) when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item examination. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the maximum limits shown on Sears Fabric Defect Scales (see 6.6) E for slubs and C for knots shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The

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sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 45 points. If one roll exceeds 45 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45 points per 100 square yards.

Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample X 3600}}{\text{Contracted width of cloth (inches) X total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur.

Baggy, ridgy, or wavy cloth
Width less than minimum specified
Poor dye penetration, mottled, streaky, or cloudy
Excessive neppiness

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

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4.2.2.4 Examination for face identification, nonidentification of preshrinkage process, and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present:

Face identification missing from either or both ends
Preshrinkage process identified by name or trademark on cloth or ticket
Not labeled in accordance with the Textile Fiber Products Identification Act

4.2.3 End item testing. The methods of testing specified in FED-STD-191 wherever applicable and as listd in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 1/4 yard full width of the dyed cloth, prior to treatment, and 6-1/2 continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Nylon identification	3.2.1.1	1530 1/
Cotton identification	3.2.1.2	1200 1/
Fiber content		
Nylon (percent)	3.2.2	2530
Cotton (percent)	3.2.2	2530 4/
Yarn (single)		
Warp	3.2.2	Visual 2/
Filling	3.2.2	Visual 2/
Labile sulfur	3.3	2020

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TABLE III. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Colorfastness to:		
Laundering (after 3 cycles)	3.3.2	5610
Crocking	3.3.2	5651
Perspiration	3.3.2	5680
Light	3.3.2	5660
Weight	3.4	5041
Yarns per inch		
Warp	3.4	5050
Filling	3.4	5050
*Tearing strength		
Filling	3.4	ASTM D 1424
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 2/
Singed	3.5	1/
Desized	3.5	1/
Mercerization	3.5	1/
Nonfibrous material	3.5.1	2611
Shrinkage		
Warp	3.6	5550
Filling	3.6	5550
Hydrostatic pressure:		
Initial	3.7	5514
After 5 launderings (cotton method)	3.7	5514, 5556
Dynamic absorption:		
Initial	3.7	5500
After 5 launderings (cotton method)	3.7	5500, 5556
Air permeability		
After 5 launderings (cotton method)	3.7	5450, 5556

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TABLE III. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Water repellents	3.7	3/
Water repellency Spray rating	3.7.1	5526
Resistance organic liquid Initial After 5 launderings (cotton method)	3.7.2 3.7.2	4.3.2 4.3.2, 5556
Cotton content	3.8	2811
Seam efficiency	3.9	5110 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination per sample unit and the result reported as "pass" or "fail".

3/ The contractor shall report the water repellents used and certify that no other material except the specified buffer (and acetic acid) has been added.

4/ The cotton content shall be calculated as follows:

$$\text{Cotton content, percent} = \frac{R}{S} \times 100$$

5/ The thread shall conform to V-T-276 and the needle shall measure 0.040 ± 0.001 inch across the blade at the eye. The needle thread shall be 40, 3 ply and 60, 3 ply for looper.

4.2.4 Packaging inspection. Examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with section 5 requirements.

4.3 Methods of inspection.

4.3.1 Test for resistance to organic liquid. Place a small specimen of the cloth on a smooth horizontal surface face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After one minute, examine the specimen under light at an angle. Absence of light reflectance at the fabric-drop interface shall be taken as evidence of the wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall be considered cause for rejection of the lot represented by the sample.

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5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth, shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B or C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use in the manufacturing of sleeping bags.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color of cloth required (see 3.3).
(c) Width of cloth required (see 3.4.1).
(d) Put-up required if other than specified (see 3.10).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Dye formulation. Satisfactory results in achieving the proper shade for Olive Green 107 and colorfastness have been obtained with an appropriate combination of the following dyes:

Cotton - Vat Green 3, CI 69500
Vat Black 25, CI 69525
Vat Green 8, CI 71050

Nylon - Neutral Premetalized Olive Green BA
Acid Brown 223
Neutral Premetalized Yellow NW

6.5 Quarpel water repellent. The "Quarpel type" water repellent treatment consists of the co-application of an emulsified fluorocarbon in combination with a fluorochemical extender. Approval of such components and combinations is the responsibility of the US Army Natick Research and Development Laboratories, Natick MA. 01760, and is based on more extensive tests including those for toxicity, which

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are not set forth in this specification. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related procurement.

6.6 Cloth defect replica. Sears Fabric Defect Scales are available from Sears, Roebuck and Co., Department 733Q, "Fabric Defect Replica Kit", Sears Tower, Chicago, IL 60684.

* 6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

* 6.8 Metric equivalents. Metric equivalents, indicated in parentheses throughout section 3 of this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.9 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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